Counter-Drone Monitoring System: Intermediate Model-Based Design and Analysis

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***Abstract*—For this midterm, I am developing a counter-drone monitoring concept for CS8395-04, AI-assisted Design of Cyber Physical Systems. The main idea is to connect the requirements, system architecture, decision logic, and physical analysis so that each result can be traced back to a design requirement. I use a generic instrument support to study the thermal and structural side of the system. Of the 512 candidate designs, 226 meet the assumed constraints. The Gaussian-process-assisted workflow selects the same lowest-mass feasible candidate using 96 physics-model evaluations, compared with 512 for the full search. Analytical comparisons, heat balance, mesh refinement, and mock supervisory tests check the numerical implementation. The current work provides the system models and an executable subsystem study. Sensor integration and physical validation are the next steps.**

***Index Terms*—model-based design, cyber-physical systems, requirements traceability, thermal analysis, structural analysis, surrogate modeling**

I. Introduction

The idea behind this project is to work through how the parts of a monitoring system fit together. The course materials use counter-unmanned-aircraft systems as a project context and explain how requirements, architecture, behavior, and parametric models describe different parts of a cyber-physical system [1], [2]. Requirements state what the system should do, the architecture assigns responsibilities, and the behavior model describes how the system responds. The physical models then connect design choices to calculated results. For the midterm, I am focusing on building those connections and showing what I can check at this stage.

The proposed system observes a defined airspace and supports human review. At this point, the project includes a documented architecture, mock supervisory logic, and a numerical model of a generic instrument support. Observation processing, field testing, real-time integration, and response hardware remain outside the implemented work. The results in this report therefore describe the progress on the modeled components. Building and validating the complete monitoring system would require those additional steps.

The main design question is how to reduce the support mass while still meeting the selected thermal and structural limits. I also want to see whether a surrogate model can find the same evaluated design with fewer calls to the physics model. This connects the course project to my interests in robotics, controls, and AI-assisted analysis because the physical design and the computational method are parts of the same engineering problem.

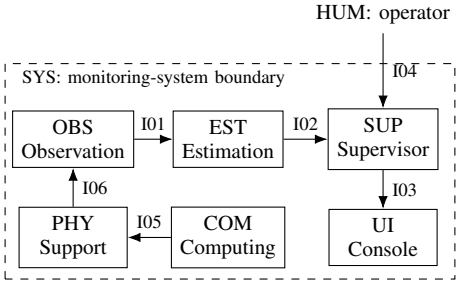


Fig. 1. System components and their interfaces. The human operator (HUM) is outside the system boundary (SYS). The implemented work covers the support analysis and mock supervisory logic.

II. System Models and Requirements

A. Architecture and Interface Allocation

Figure 1 shows the system boundary and what each component is responsible for. Observation (OBS), estimation (EST), and the user interface (UI) are specified in the architecture. The supervisor (SUP) contains the mock decision rules. The physical support (PHY) is the part analyzed numerically, and the computing component (COM) supplies an assumed heat load.

The six connections in the diagram are also recorded in model/system_model.json. I01 sends the source timestamp, health, and confidence from OBS to EST. I02 sends observation availability and confidence from EST to SUP. I03 sends the mode, mock alert, and health status to UI, while I04 provides the operator's review input. On the physical side, I05 specifies the heat load in watts from COM to PHY, and I06 describes how OBS mounts to PHY. The reference audit checks that the connections point to the correct components and use consistent types. Actual message delivery and timing still need to be tested.

B. Supervisory Behavior

The decision logic in Fig. 2 follows a specific order: check sensor health, check whether an observation is available, and then check whether an operator has reviewed it. An unhealthy input puts the system in Degraded mode. A healthy

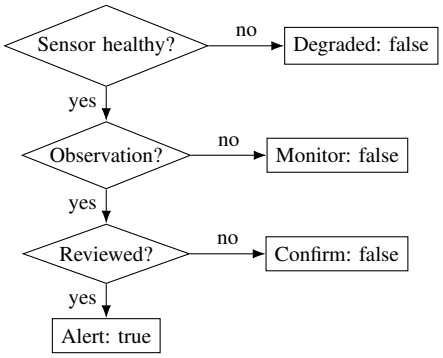


Fig. 2. Order of the supervisory checks. Each output shows the mode and mock alert flag. The current logic is stateless; persistent state and timing still need to be implemented.

input without an observation produces Monitor mode. If an observation is available but has not been reviewed, the output is Confirm. A reviewed observation produces Alert, which is the only branch that sets the mock alert flag.

V-FAULT, V-REVIEW, and V-MODE-SEQUENCE test the function src/physics.py:supervise using representative mock inputs. These checks show whether the code follows the intended decision rules. Detection accuracy and the response time of the integrated system require separate tests with actual observation processing and connected components.

C. Requirements and Evidence Links

Table I puts all eleven requirements next to the responsible components and the evidence available so far. The file model/requirements_evidence.json keeps the full requirement wording, source references, calculated results, and tests that remain open. The physical acceptance limits are assumptions for this demonstration and will need to be revisited when representative physical data are available.

For example, R04 assigns the temperature limit to PHY. Relation P01 and the thermal solver calculate the temperature, and the selected candidate provides the value used to check that requirement. The analytical comparison and heat-balance check help verify the calculation. Physical temperature measurements are still required under R11. Requirements R05–R08 follow the same idea: start with geometry and material assumptions, calculate the structural response, and connect the result to the requirement and its verification evidence.

III. Parametric and Physical Analysis Models

A. Shared Parameters and Assumptions

I vary two dimensions: support length $L \in [0.12, 0.24]$ m and thickness $t \in [0.002, 0.008]$ m. Table II lists the other inputs, which are held fixed for the study. For the rectangular support, the area is $A = bt$, the second moment of area is $I = bt^3/12$, the exposed perimeter is $p = 2(b + t)$, and the support mass is $m_s = \rho btL$.

Figure 3 shows how the physical calculations connect. The six relations P01–P06 in model/parametric_relations.json cover thermal response, temperature-dependent modulus, static deflection, first natural frequency, root stress, and geometry/mass. The thermal model runs first. Its temperature results then change the assumed modulus used by the structural model. This is a one-way connection between the analyses, and the current implementation does not iterate between them.

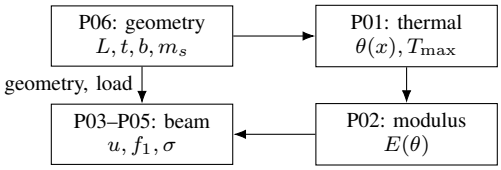


Fig. 3. Connections between the geometry, thermal, and structural calculations. Temperature changes the assumed modulus used in the beam model, and support mass is the design objective.

B. Thermal and Structural Formulation

For the thermal calculation, I define $\theta = T - T_a$ and use the following one-dimensional fin model:

$$(kA\theta')' - hp\theta = 0, \quad \theta(0) = 0, \quad kA\theta'(L) = Q. \tag{1}$$

The clamp stays at ambient temperature, and heat enters at the tip. With constant properties, the analytical tip-temperature rise is

$$\theta(L) = \frac{Q \tanh(\mu L)}{kA\mu}, \qquad \mu = \sqrt{\frac{hp}{kA}}. \tag{2}$$

The finite-element temperature solution then sets the modulus for each element using the assumed relationship

$$E(\theta) = E_0(1 - \beta\theta). \tag{3}$$

The structural calculation uses an Euler–Bernoulli beam model [3] to solve

$$K(T)u = F, \qquad K(T)\phi = \omega^2 M\phi, \qquad f_1 = \frac{\omega_1}{2\pi}. \tag{4}$$

The load vector includes the tip payload and the support's own weight. The consistent beam mass matrix also includes the payload mass. Root bending stress comes from the root moment and the section geometry. These calculations assume linear behavior, small deflections, and the stated boundary conditions. A detailed three-dimensional support analysis and physical validation remain future work.

IV. Numerical Verification

Before comparing designs, I check the physics calculations against independent analytical results. The script src/verify.py also checks heat balance and the effect of mesh refinement. Table III gives the results. The mesh check compares 24-element results with a 64-element reference at $L = 0.18$ m and $t = 0.004$ m. The static analytical check uses constant-modulus formulas for the tip load and self-weight, while the modal check uses the characteristic root for an unloaded cantilever. All errors in the table are relative differences except the heat residual, which is reported in watts. The seven numerical checks and four mock/invalid-input checks pass,

TABLE I
REQUIREMENT ALLOCATION AND INTERMEDIATE EVIDENCE

ID	Owner	Requirement	Evidence and current status
R01	SYS	Monitoring and operator-review boundary	Architecture and scope review; descriptive baseline.
R02	SUP	Unhealthy input blocks a mock alert	V-FAULT: Degraded mode and alert false; mock logic verified.
R03	SUP	Unreviewed observation blocks a mock alert	V-REVIEW: Confirm mode and alert false; mock logic verified.
R04	PHY	Maximum temperature $\leq 65\,^{\circ}\mathrm{C}$	T-TEMP: $64.454\,^{\circ}\mathrm{C}$; numerically satisfied.
R05	PHY	Tip deflection ≤ 0.5 mm	T-DEFLECTION: 0.3436 mm; numerically satisfied.
R06	PHY	First natural frequency ≥ 22 Hz	T-FREQUENCY: 27.330 Hz; numerically satisfied.
R07	PHY	Root stress ≤ 40 MPa	T-STRESS: 5.073 MPa; numerically satisfied.
R08	PHY	Support reach ≥ 0.16 m	T-REACH: 0.161613 m; numerically satisfied.
R09	SYS	Reject out-of-domain surrogate predictions	T-DOMAIN: recorded rejection check passed.
R10	OBS	Timestamp, health, and confidence contract	T-OBS-CONTRACT: integration test pending.
R11	PHY	Independent physical measurements	T-PHYSICAL: experimental validation pending.

TABLE II
ASSUMED PHYSICAL-SUBSYSTEM INPUTS

Parameter	Value
Width b; tip payload	0.04 m; 0.25 kg
Heat input Q; ambient T_a	8 W; $25\,^{\circ}\mathrm{C}$
Convection h	12 W/(m^2 K)
Conductivity k; density ρ	167 W/(m K); 2700 kg/m^3
Reference modulus E_0	69 GPa
Temperature coefficient β	0.0004 K^{-1} (uncalibrated)
Thermal / beam discretization	24 / 24 elements

TABLE III
NUMERICAL VERIFICATION RESULTS

Check	Observed error	Limit
Thermal analytical	4.85×10^{-5}	2×10^{-3}
Heat balance (W)	1.79×10^{-13}	10^{-8}
Static analytical	5.44×10^{-12}	10^{-6}
Bare-beam first mode	2.47×10^{-8}	10^{-3}
Mesh: temperature	2.53×10^{-5}	2×10^{-3}
Mesh: deflection	5.69×10^{-6}	2×10^{-3}
Mesh: frequency	2.84×10^{-6}	2×10^{-3}

giving eleven recorded checks in results/verification.json. The separate surrogate domain check also passes. This shows that the implementation is consistent with the reference calculations at the tested conditions. Physical validation still requires measurements of material behavior, mounting compliance, convection, and field loads. The course's dynamics and fidelity material makes this same distinction between checking the numerical implementation and checking how well it represents the physical system [4].

V. DESIGN EXPLORATION AND INTERMEDIATE RESULTS

A. Study Protocol

I use Latin-hypercube sampling [5] to create 64 training points, 64 validation points, 192 final held-out test points, and 512 candidate points. The corresponding seeds are 11, 22, 55, and 44. Quadratic/Ridge, thin-plate radial basis function (RBF), and Gaussian process (GP) surrogates all fit the same training data. The GP uses a constant-times-ARD-RBF kernel, normalized log targets, $\alpha = 10^{-8}$, and one optimizer restart [6]. The bounded Powell optimizer uses training likelihood only, and the code checks that the optimization finishes successfully [7].

An earlier pilot split used seed 33 during development. A check using the training data led to a correction to the optimizer. The fresh final split was evaluated after that correction, with no further tuning afterward. Keeping that order matters because the final test set needs to stay separate from model development. The final results show how closely the surrogates match the reduced physics model within the design domain. Comparison with physical measurements remains a later step.

B. Constrained Support Selection

Of the 512 candidates, 226 meet all the assumed limits. The GP-assisted method uses prediction intervals to screen the candidates, ranks the predicted feasible designs by mass, and checks up to 32 of them with the physics model. During selection, it does not have access to the physics results from the full candidate search. The full search provides the separate comparison for judging the assisted method.

Both methods select the same lowest-mass feasible candidate, shown in Table IV, so the mass gap is zero. This tells me that the assisted method recovers the full search's choice within the evaluated set. A search over these candidates does not establish the best possible design across the entire continuous domain. The selected temperature is also close to

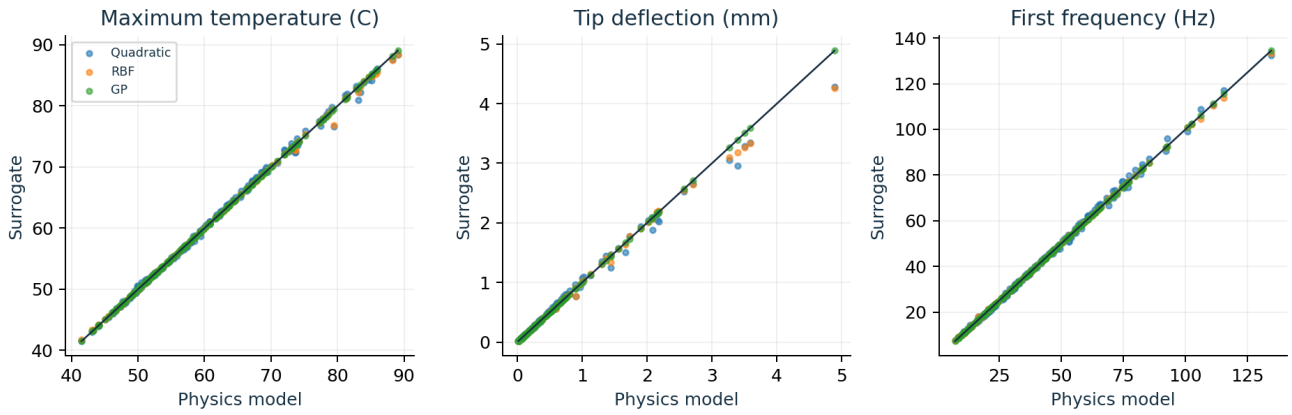


Fig. 4. Surrogate predictions compared with the reduced physics model for the same 192 held-out test designs. Each panel shows one output. The plots check interpolation within the assumed design domain.

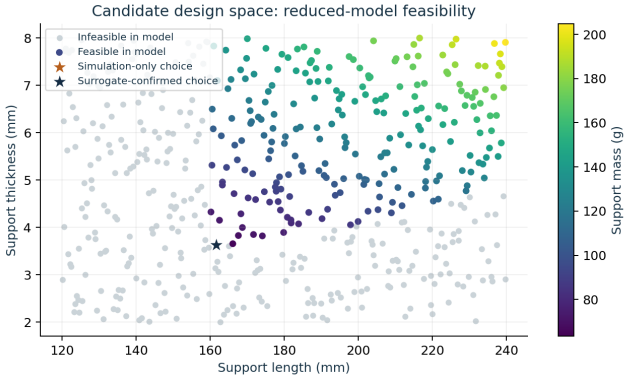


Fig. 5. Candidate support designs and the feasible region predicted by the reduced model. The full search and GP-assisted method select the same candidate.

TABLE IV
SELECTED CANDIDATE AND ASSUMED ACCEPTANCE LIMITS

Quantity	Model result	Limit
Length	161.613 mm	$\geq$ 160 mm
Thickness	3.633 mm	design variable
Support mass	63.415 g	minimize
Maximum temperature	64.454 °C	$\leq$ 65 °C
Tip deflection	0.3436 mm	$\leq$ 0.5 mm
First natural frequency	27.330 Hz	$\geq$ 22 Hz
Root stress	5.073 MPa	$\leq$ 40 MPa

the assumed limit, which makes measured thermal boundary conditions a priority before moving the design forward.

C. Surrogate Accuracy and Evaluation Cost

Table V compares the final held-out root-mean-square error (RMSE), and Fig. 4 shows the corresponding predictions. The GP has the smallest error for all three outputs in this test. Its nominal 95% prediction intervals contain 95.31% of the temperature results and 92.71% of both structural

TABLE V
FINAL HELD-OUT RMSE, 192 DESIGNS

Model	Temperature (C)	Deflection (mm)	Frequency (Hz)
Quadratic	0.43997	0.070514	0.97140
RBF	0.24469	0.058014	0.25904
GP	0.00400	0.000409	0.00862

outputs. The structural coverage is below the nominal level, so these intervals cannot be treated as guaranteed bounds. The workflow therefore keeps the physics-model confirmation step when selecting a design.

The main computational result is the reduction from 512 physics evaluations to 96: 64 for training and 32 for confirmation. This is an 81.25% reduction in model calls, and all 32 confirmed candidates meet the assumed limits. On the final test set, the GP interval screen accepts 85 points with no false-feasible or false-infeasible classifications. Those results apply to the tested points; reliability beyond the tested domain still needs to be established.

I also timed both workflows across three trials. The recorded median was 0.385 s for the full search and 0.311 s for the GP workflow. These times include the simulation, fitting, prediction, and confirmation work within each method. Validation, held-out testing, mesh checks, warm-up, and plotting are additional offline costs excluded from both workflow totals. The timing difference is small and depends on the runtime environment. The reduction in physics-model calls is the more useful efficiency result for this study.

VI. MIDTERM STATUS AND COURSE ALIGNMENT

A. Completed Artifacts and Remaining Work

For the midterm, the completed work includes the scope baseline (A01), block/interface architecture (A02), supervisory activity model (A03), requirements-to-evidence matrix (A04), parametric relations (A05), numerical verification (A06), support design study (A07), and surrogate comparison (A08).

Together, these files show how the design assumptions lead to the reported results and how those results connect to the requirements. The JSON records and diagrams describe the model-based design. Formal SysML conformance has not been established.

The two requirements still open are R10 and R11. The next step for the system is to implement the timestamp, health, and confidence contract, connect the console and operator-review input, and test faults and timing across the interfaces. The physical study needs representative CAD, calibrated material properties, measured mounting stiffness, and measured thermal boundary conditions. Independent measurements of temperature, static deflection, and natural frequency would then show where the model agrees with the hardware and where it needs correction. A revised surrogate would also need recalibration and a new reserved final test set.

B. Connection to Research and the Course

What I am trying to build through this project is a way to connect the controls, robotics, physical modeling, and AI parts of a design problem. The requirements set the acceptance criteria, the interfaces describe how the components connect, the physical models calculate performance, and the surrogates reduce repeated analysis work. These pieces support the same design process. I could apply that process to instrument mounts or robotic subsystems by replacing the current assumptions with the appropriate models and measurements. Applying it to friction-stir welding would also require welding-process physics and experimental data.

This also follows the progression of the course. The MBSE lectures connect requirements with structural, behavioral, and parametric views [1]. Architectural modeling adds typed interfaces and checks that the references are consistent [2]. Design-space exploration provides the variables, constraints, and candidate comparisons used in the support study [8]. The dynamics and higher-fidelity analysis material connects to the thermal and modal checks and to the distinction between numerical verification and physical validation [4], [9]. Finally, the multidisciplinary optimization material connects to the shared analysis dependencies and the surrogate comparison [10]. The current implementation passes thermal results into the structural analysis in one direction. Full coupled optimization remains future work.

VII. Conclusion

For this midterm, I have connected the system models to an executable support analysis and recorded evidence for the requirements. The reduced model passes the numerical checks and identifies a 63.415-g support candidate that meets the assumed limits. The GP-assisted method finds the same evaluated candidate with 81.25% fewer physics calls. The next step is to replace the main physical assumptions with measurements and complete the observation-to-review integration. As that work develops, the requirement-to-evidence links need to stay connected to the updated models and results.

Appendix A
Artifact Access and Reproduction

The full analysis archive keeps the system models in model/, the physical and verification code in src/, the inputs in config.json, and the recorded results in results/. For Overleaf, the editable report is main.tex. The package also includes the IEEEtran class and the report figures in figures/. The analysis archive's README.md explains the environment and run order.

To reproduce the full study, install the packages in requirements.txt and run `python reproduce.py`. The recorded environment uses Python 3.12 and a LaTeX installation with pdflatex. To rebuild the report from the recorded results, run `python build_report.py`. The result files keep the full numerical precision, while the tables use rounded values. The figure-generation code and verification records are included so the results can be checked.

References

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[8] CS8395-04, supplied lecture L06, design-space exploration, pp. 8–10, 21, 39.

[9] CS8395-04, supplied lecture L08, higher-fidelity analysis, pp. 3–6, 9, 14–15.

[10] CS8395-04, supplied lecture L09, multidisciplinary analysis and design optimization, pp. 10–18, 24–27.